MD OMAR FARUK MARUF

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ACADEMIC / RESEARCH INTEREST

Computer Vision, High-Performance Cyber-Physical Systems, Blockchain Architecture, Applied Cybersecurity, Machine Learning, and Autonomous Systems.

EDUCATION

University of Global Village (UGV) — Barishal, Bangladesh
Bachelor of Science in Computer Science and Engineering (Feb 2020 – Feb 2024)

CGPA: 3.69 / 4.00 (Last Semester CGPA: 3.80 / 4.00)
Relevant Coursework: Artificial Intelligence, Machine Learning, Pattern Recognition, Algorithm Design & Analysis, Data Structures, Parallel & Distributed Processing, Microprocessor & Assembly Language, Computer Architecture, VLSI Design, Discrete Mathematics, Mathematics I–II

PUBLICATIONS

Maruf, MD Omar Faruk, Md Riadul Islam, Md Jubayar Alam Rafi. "Non-permissible Mobile Detection to Enhance Security in Bangladeshi Museum: A Multiprocess YOLO-Based Approach." In *6th International Conference on Electrical Engineering and Information & Communication Technology (ICEEICT)*, pp. 37-42. IEEE, 2024.
DOI: 10.1109/ICEEICT62016.2024.10534325

RESEARCH & PROFESSIONAL EXPERIENCE

Software Engineer | SMC Labs BD (June 2025 – Present)

- Lead research and development on blockchain core architecture design, modified Bitcoin wallet codebases, and protocol optimizations using advanced C++.
- Conduct rigorous open-source codebase analysis (Bitcoin) to implement custom cryptographic and security features for next-generation blockchain infrastructure.
- Field of expertise: C++, Blockchain, dApp, Solidity, NextJS, NestJS, JavaScript,, HTML & CSS

Software Intern (Research & Development) | SMC Labs BD (Dec 2024 – May 2025)

- Collaborated with senior software engineers on open-source Bitcoin core research, decentralization technologies, and system performance tuning.
- Authored detailed technical documentation and research specifications for enterprise dApp integration.

Python Trainer | Sheikh Russel School of Future, ICT Division (May 2024 – Oct 2024)

- Instructed secondary school students (Classes 6–10) in foundational computer science, algorithm development, and data structures using Python.
- Prepared and mentored students for national robotics competitions and government-funded tech initiatives.

PROJECTS

AGAAW | Peer-to-Peer Academic Mentorship Platform (www.agaaw.com)

- Designed and deployed an international mentorship engine linking prospective graduate students with global research mentors.

- Integrated real-time video conferencing, dynamic chat architecture, scholarship search algorithms, and geographical discovery tools.

HONORS & AWARDS

Champion of imaGen Ventures Youth Challenge 2023, (23 October 2023)
UNICEF Bangladesh, Generation Unlimited
200 teams had applied and only 11 teams were selected. We developed an automatic robot system "SYNERGY" which can detect and collect plastic wastes more efficiently (25% faster by implementing multi-process)

- We secured the Champion title and won $1000 prize money.
- 3 days of training on the Art of Problem Definition.

Champion: Smart Entrepreneurs in District 2025, (19 October 2024)
GP Accelerator, Grameenphone Ltd

We proposed an Emergency Response Device idea with a prototype to address rising river water levels and fluctuations in pH and salinity, which pose flooding risks and impact agriculture.

- We secured the Champion title among the 17 different ideas and won 1,00,000 BDT prize money.

TECHNICAL SKILLS

- **Languages:** C++ (Advanced), Python (Advanced), Solidity, JavaScript
- **Core Domains:** Computer Vision, Deep Learning, Blockchain Infrastructure, Embedded Systems, Real-time Systems
- **Hardware & IoT:** Raspberry Pi, NVIDIA Jetson Nano, Microcontrollers
- **Frameworks & Tools:** NextJS, NestJS, OpenCV, Git/GitHub, Linux Environments

LEADERSHIP & EXTRACT-CURRICULAR ACTIVITIES

President | DYDF, Babuganj Branch (Nov 2024 – Present)

- Lead district-level youth empowerment, educational literacy programs, and community technology awareness campaigns.

Active Volunteer Member | Volunteer for Bangladesh (VBD) (Oct 2023 – Jan 2025)

- Organized emergency disaster response and relief operations for flood-affected populations.

CERTIFICATIONS & STANDARDIZED TESTS

- **IELTS Academic:** Overall Band 6.5 (Speaking: 6.5, Listening: 6.5, Writing: 6.5, Reading: 6.0)
- **Coursework:** Introduction to Cybersecurity | Advance Networking | C++ | Python
- **Languages:** Bengali (Native), English (Fluent), German (A1 Basic)

ACADEMIC REFERENCES

1. Md Riadul Islam
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Department of Computer Science
University of Global Village (UGV)
Email: riadul.cse@ugv.edu.bd
Phone: +8801842611852

2. Fazle Rabbi
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